12

Heat sterilisation

Heat sterilisation is the unit operation in which foods are heated at a sufficiently high temperature and for a sufficiently long time to destroy microbial and enzyme activity. As a result, sterilised foods have a shelf life in excess of six months at ambient temperatures. The severe heat treatment during the older process of in-container sterilisation (canning) may produce substantial changes in nutritional and sensory qualities of foods. Developments in processing technology therefore aim to reduce the damage to nutrients and sensory components, by either reducing the time of processing in containers or processing foods before packaging (aseptic processing). More recent developments, including ohmic heating, are described in Chapter 18. The theory of thermal destruction of micro-organisms and the effect of heat on nutrients and sensory components of foods is described in Chapter 1. In this chapter the effects of microbial heat resistance on the design of heat sterilisation procedures and equipment are described, first for in-container heat sterilisation and then for ultra high-temperature (UHT) processes.

12.1 In-container sterilisation

12.1.1 Theory

The length of time required to sterilise a food is influenced by:

- the heat resistance of micro-organisms or enzymes likely to be present in the food
- the heating conditions
- the pH of the food
- the size of the container
- the physical state of the food.

In order to determine the process time for a given food, it is necessary to have information about both the heat resistance of micro-organisms, particularly heat resistant spores, or enzymes that are likely to be present and the rate of heat penetration into the food.

Heat resistance of micro-organisms

The factors that influence heat resistance of micro-organisms or enzymes and their characterisation by D and z values are described in Chapter 1 (Section 1.4.5). Most heat resistant spores have a z value of around 10ºC. This temperature rise will result in a ten-fold reduction in processing time needed to achieve the same lethality. In low-acid foods (pH > 4.5), the heat resistant, spore forming micro-organism, *Clostridium botulinum* is the most dangerous pathogen likely to be present. Under anaerobic conditions inside a sealed container it can grow to produce a powerful exotoxin, botulin, which is sufficiently potent to be 65% fatal to humans. *Cl. botulinum* is ubiquitous in soil and it is therefore likely to be found in small numbers on any raw material that has contact with soil. Because of the extreme hazard from botulin, the destruction of this micro-organism is therefore a *minimum* requirement of heat sterilisation. Normally, foods receive more than this minimum treatment as other more heat-resistant spoilage bacteria may also be present (Table 12.1). In more acidic foods (pH 4.5–3.7), other micro-organisms (for example yeasts and fungi) or heat-resistant enzymes are used to establish processing times and temperatures. In acidic foods (pH < 3.7), enzyme inactivation is the main reason for processing and heating conditions are less severe (sometimes referred to as *pasteurisation*).

Thermal destruction of micro-organisms takes place logarithmically (Chapter 1) and a sterile product cannot therefore be produced with certainty no matter how long the process time. However, the *probability* of survival of a single micro-organism can be predicted using details of the heat resistance of the micro-organism and the temperature and time of heating. This gives rise to a concept known as *commercial sterility*. For example, a process that reduces cell numbers by twelve decimal reductions (a 12D process), applied to a raw material which contains 1000 spores per container would reduce microbial numbers to 10^{-9} per container, or the probability of one microbial spore surviving in one billion containers processed. Commercial sterility means in practice that

Table 12.1 Heat resistance of some spore-forming bacteria[a] used as a basis for heat sterilisation of low acid foods

Micro-organism	z value (ºC)	D_{121} value (min)	Typical foods
Thermophilic (35–55ºC)			
Bacillus stearothermophilus	9–10	3.0–4.0	Vegetables, milk
Clostridium thermosaccharolyticum	7.2–10	3.0–4.0	Vegetables
Mesophilic (10–40ºC)			
Clostridium sporogenes	8.8–11.1	0.7–1.5	Meats
Bacillus subtilis	4.1–7.2	0.3–0.76	Milk products
Cl. botulinum toxins A and B	5.5	0.1–0.3	Low-acid foods
B. coagulans	6–9	0.01–0.07	Milk
B. cereus	36	3.8	Milk
Psychrophilic (−5−1.5ºC)	10	3.0 (60ºC)	Low-acid foods
Cl. botulinum toxin E			

[a] Note: the data is intended to be indicative only as the thermal resistance of micro-organisms is influenced by the nature of the food. Original literature gives precise information for particular products.
Adapted from Lund (1975), Burton (1988), Brennan *et al.* (1990), Heldman and Hartel (1997) and Licciardello *et al.* (1967).

the heat processing inactivates substantially all micro-organisms and spores which, if present, would be capable of growing in the food under defined storage conditions (Brennan *et al*., 1990).

The level of survival is determined by the type of micro-organism that is expected to contaminate the raw material. A 12*D* process is used when *C. botulinum* is likely to be present in low acid foods, but in foods that contain more heat-resistant spoilage micro-organisms (Table 12.1), the application of a 12*D* process would result in over-processing and excessive loss of quality. In practice a 2*D* to 8*D* process is therefore used to give the most economical level of food spoilage consistent with adequate food quality and safety. However, because of the lower heat resistance of *C. botulinum*, the probability of survival remains similar to that obtained in a 12*D* process. For these processes to operate successfully, the microbial load on raw materials must be kept at a low level by hygienic handling and preparation procedures (Chapter 3), and in some foods by blanching (Chapter 10). Any failure in these procedures would increase the initial numbers of cells and, because of the logarithmic rate of destruction, would increase the incidence of spoilage after processing. In canning factories, accelerated storage trials on randomly selected cans of food ensure that these levels of commercial sterility are maintained before foods are released for retail sale. When *Cl. botulinum* grows and produces toxin in a sealed container there is characteristic production of gas which can cause visible swelling of the container (although this is not the only cause of swelling). Routine quality assurance measures therefore include observation for swollen or bloated cans.

In addition to information on heat resistance, it is necessary to collect data describing the rate of heat penetration into the food in order to calculate the processing time needed for commercial sterility.

Rate of heat penetration

Heat is transferred from steam or pressurised water through the container and into the food. Generally the surface heat transfer coefficient (Chapter 1) is very high and is not a limiting factor in heat transfer. The following factors are important influences on the rate of heat penetration into a food:

- *Type of product*. Liquid or particulate foods (for example peas in brine) in which natural convection currents are established heat faster than solid foods in which heat is transferred by conduction (for example meat pastes and corned beef) (Fig. 12.1). The low thermal conductivity of foods is a major limitation to heat transfer in conduction heating foods.
- *Size of the container*. Heat penetration to the centre is faster in small containers than in large containers.
- *Agitation of the container*. End-over-end agitation (Fig. 12.2) and, to a lesser extent, axial agitation increases the effectiveness of natural convection currents and thereby increases the rate of heat penetration in viscous or semi-solid foods (for example beans in tomato sauce).
- *Temperature of the retort*. A higher temperature difference between the food and the heating medium causes faster heat penetration.
- *Shape of the container*. Tall containers promote convection currents in convective heating foods.
- *Type of container*. Heat penetration is faster through metal than through glass or plastics owing to differences in their thermal conductivity (Table 1.5).

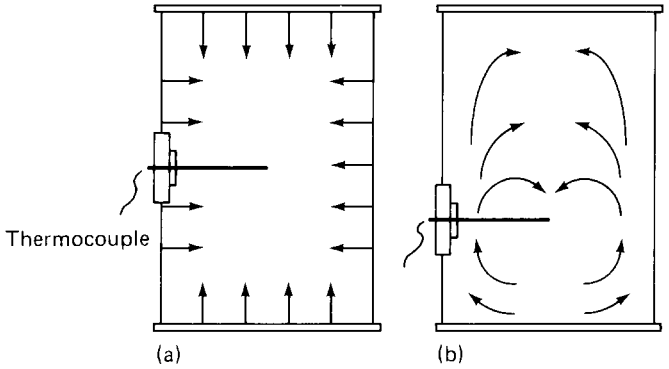


Fig. 12.1 Heat transfer into containers by (a) conduction and (b) convection.

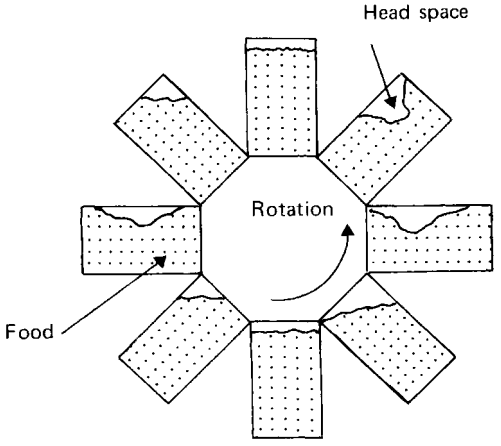


Fig. 12.2 End-over-end agitation of containers.
(After Hersom and Hulland (1980).)

The rate of heat penetration is measured by placing a thermocouple at the thermal centre of a container (the point of slowest heating) to record temperatures in the food during processing (Fig. 12.3). It is assumed that all other points in the container receive more heat and are therefore adequately processed. In continuous sterilisers, self-contained miniature temperature recorders–transmitters can be placed at the thermal centre of the container to measure heat penetration.

In cylindrical containers the thermal centre is at the geometric centre for conductive heating foods and approximately one third up from the base of the container for convective heating foods (Fig. 12.1). However, in convective heating, the exact position varies and should be found experimentally. Convective heating is more rapid than conductive heating and the rate depends mostly on the viscosity of the food. In commercial processing, containers of viscous food may be agitated to increase the rate of convective heating (Section 12.1.3).

A typical heating curve is produced by plotting temperature versus time on semilogarithmic graph paper (Fig. 12.4). A broken heating curve occurs when a food is initially heated by convective heating but then undergoes a rapid transition to conductive heating (for example in foods which contain a high concentration of starch which undergoes a sol-to-gel transition).

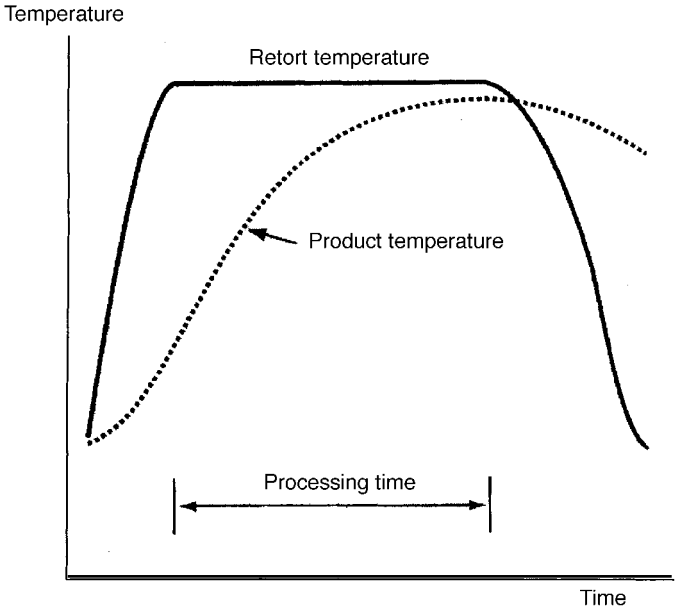


Fig. 12.3 Heat penetration into a can of conductive heating food: (a) = retort temperature; (b) = temperature at the slowest heating point.

The thermal death time (TDT), or *F value*, is used as a basis for comparing heat sterilisation procedures. It is the time required to achieve a specified reduction in microbial numbers at a given temperature and it thus represents the total time–temperature combination received by a food. It is quoted with suffixes indicating the retort temperature and the *z* value of the target micro-organism. For example, a process operating at 115ºC based on a micro-organism with a *z* value of 10ºC would be expressed as F_{115}^{10}. The *F* value may also be thought of as the time needed to reduce microbial numbers by a multiple of the *D* value. It is found using:

$$F = D\,(\log n_1 - \log n_2) \qquad 12.1$$

where n_1 = initial number of micro-organisms and n_2 = final number of micro-organisms.

A reference *F* value (F_0) is used to describe processes that operate at 121ºC which are based on a micro-organism with a *z* value of 10ºC. Typical F_0 values are 3–6 min for vegetables in brine, 4–5 min for cream soups and 12–15 min for meat in gravy.

Calculation of process times

The slowest heating point in a container may not reach the processing temperature (Fig. 12.3), but once the temperature of the food rises above approximately 70ºC, thermal destruction of micro-organisms takes place. The purpose of calculating processing time is to enable a retort operator to know how long a food in a given can size should be held at a set processing temperature in order to achieve the required thermal destruction at the slowest heating point in the container. Two methods are described in the following section, the first being a mathematical method based on the equivalent lethality of different time–temperature combinations and the second being a graphical method. Both are discussed in detail by Stumbo (1973) and Ball and Olson (1957) who also describe a rapid method based on the use of nomograms. Holdsworth (1997) compares a number of other mathematical methods.

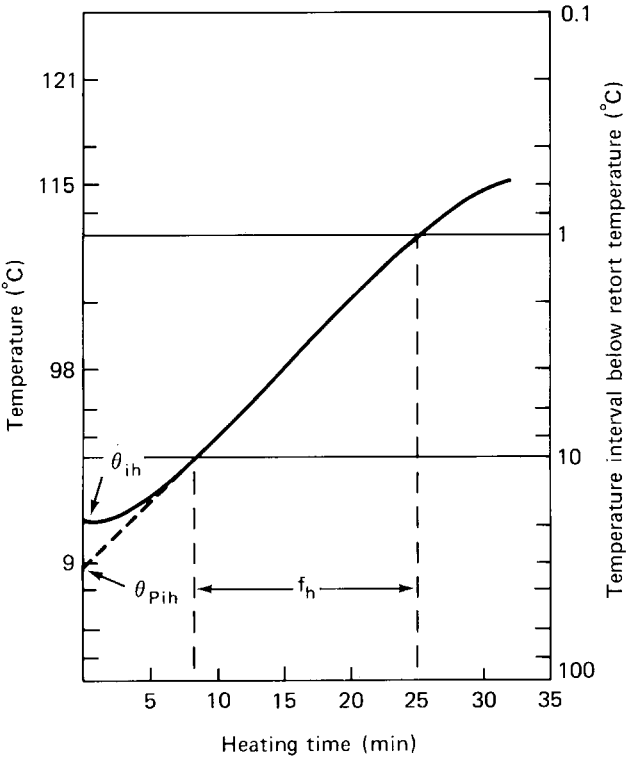


Fig. 12.4 Heating curve.

Formula (or mathematical) method

This method enables rapid calculation of process times for different retort temperatures or container sizes, but is limited by the assumptions made about the nature of the heating process. The method is based on:

$$B = f_h \log \left(\frac{j_h I_h}{g} \right) \qquad 12.2$$

where B (min) = time of heating, f_h (min), the heating rate constant = the time for the heat penetration curve to cover one logarithmic cycle, j_h = the thermal lag factor found by extrapolating the curve in Fig 12.4 to find the pseudo-initial product temperature (θ_{pih}).

$$j_h = \frac{\theta_r - \theta_{pih}}{\theta_r - \theta_{ih}} \qquad 12.3$$

where I_h ($= \theta_r - \theta_{ih}$) (°C) = the difference between the retort temperature and the initial product temperature, g = the difference between the retort temperature and the final product temperature at the slowest heating point, θ_r (°C) = retort temperature and θ_{ih} (°C) = initial product temperature.

The heating rate constant varies according to the surface area:volume ratio of the container and therefore depends on the shape and size of the pack. It also depends on whether the product heats by convection or conduction. With the exception of g the above information can be found from the heating curve (Fig. 12.4). Cooling rate data are plotted in a similar way.

The value of g is influenced by the following factors:

- the TDT of the micro-organism on which the process is based
- the slope f_h of the heating curve
- the z value of the target micro-organism
- the difference between the retort temperature and the temperature of the cooling water.

To take account of these variables, Ball (1923) developed the concept of comparing the F value at the retort temperature (denoted F_1) with a reference F value of 1 min at 121ºC (denoted F). The TDT at the retort temperature is described by the symbol U and is related to the reference F value and F_1 using:

$$U = FF_1 \qquad 12.4$$

If the reference F value is known, then it is possible to calculate U by consulting F_1 tables (Table 12.2). The value of g may then be found from f_h/u and g tables (Table 12.3).

One further factor which influences the value of g is the difference between the retort temperature and the temperature of the cooling water. In conductive heating foods, there is a lag before cooling water begins to lower the product temperature, and this results in a significant amount of heating after the steam has been turned off. It is therefore necessary to include a cooling lag factor j_c. This is defined as the 'time taken for the cooling curve to cover one logarithmic cycle', and is analogous to j_h, the heating lag factor. The cooling portion of the heat penetration curve is extrapolated to find the pseudo-initial product temperature θ_{pic} at the start of cooling, in a similar way to θ_{pih}. j_c is found using:

$$j_c = \frac{\theta_c - \theta_{pic}}{\theta_c - \theta_{ic}} \qquad 12.5$$

where θ_c (ºC) = cooling water temperature and $-\theta_{ic}$ (ºC) = the actual product temperature at the start of cooling. When using Table 12.3, the appropriate value of j_c can then be used to find g.

Finally, in batch retorts, only 40% of the time taken for the retort to reach operating temperature (the come-up time l) is at a sufficiently high temperature to destroy micro-organisms. The calculated time of heating (B) is therefore adjusted to give the corrected processing time:

$$\text{process time} = \text{B} - 0.4l \qquad 12.6$$

Table 12.2 F_1 values for selected z values at retort temperatures below 121ºC

$121-\theta_r$ (ºC)	z value					
	4.4ºC	6.7ºC	8.9ºC	10ºC	11.1ºC	12ºC
5.6	17.78	6.813	4.217	3.594	3.162	2.848
6.1	23.71	8.254	4.870	4.084	3.548	3.162
6.7	31.62	10.00	5.623	4.642	3.981	3.511
7.2	42.17	12.12	6.494	5.275	4.467	3.899
7.8	56.23	14.68	7.499	5.995	5.012	4.329
8.3	74.99	17.78	8.660	6.813	5.623	4.806
8.9	100.0	21.54	10.00	7.743	6.310	5.337
9.4	133.4	26.10	11.55	8.799	7.079	5.926
10.0	177.8	31.62	13.34	10.00	7.943	6.579
10.6	237.1	38.31	15.40	11.36	8.913	7.305

Adapted from Stumbo (1973).

Table 12.3 Selected f_h/U and g values when z = 10 and j_c = 0.4–2.0

	Values of g for the following j_c values					
f_h/U	0.40	0.80	1.00	1.40	1.80	2.00
0.50	0.0411	0.0474	0.0506	0.0570	0.0602	0.0665
0.60	0.0870	0.102	0.109	0.123	0.138	0.145
0.70	0.150	0.176	0.189	0.215	0.241	0.255
0.80	0.226	0.267	0.287	0.328	0.369	0.390
0.90	0.313	0.371	0.400	0.458	0.516	0.545
1.00	0.408	0.485	0.523	0.600	0.676	0.715
2.00	1.53	1.80	1.93	2.21	2.48	2.61
3.00	2.63	3.05	3.26	3.68	4.10	4.31
4.00	3.61	4.14	4.41	4.94	5.48	5.75
5.00	4.44	5.08	5.40	6.03	6.67	6.99
10.0	7.17	8.24	8.78	9.86	10.93	11.47
20.0	9.83	11.55	12.40	14.11	14.97	16.68
30.0	11.5	13.6	14.6	16.8	18.9	19.9
40.0	12.8	15.1	16.3	18.7	21.1	22.3
50.0	13.8	16.4	17.7	20.3	22.8	24.1
100.0	17.6	20.8	22.3	25.4	28.5	30.1
500.0	26.0	30.6	32.9	37.5	42.1	44.4

Adapted from Stumbo (1973).

More complex formulae are necessary to calculate processing times where the product displays a broken heating curve.

Sample problem 12.1
A low-acid food is heated at 115ºC using a process based on $F_{121.1}^{10} = 7$ min. From heat penetration data the following information was obtained: $\theta_{ih} = 78$ºC, $f_h = 20$ min, $j_c = 1.80$, $f_c = 20$ min, $\theta_{pih} = 41$ºC and $\theta_{ih} = 74$ºC. The retort took 11 min to reach process temperature. Calculate the processing time.

Solution to Sample problem 12.1
From equation (12.3),

$$j_h = \frac{115 - 41}{115 - 74}$$
$$= 2.00$$

and

$$I_h = 115 - 78$$
$$= 37\text{ºC}$$

From Table 12.2 (for $121.1 - \theta_r = 6.1$ and $z = 10$ºC),

$$F_1 = 4.084$$

From equation (12.4),

$$U = 7 \times 4.084$$
$$= 28.59,$$

$$\frac{f_h}{U} = \frac{20}{28.59}$$
$$= 0.7$$

From Table 12.3 (for $f_h/U = 0.7$, $j_c = 1.80$),

$$g = 0.241^\circ\text{C}$$

(that is the thermal centre reaches 114.76ºC). From equation (12.2),

$$B = 20 \log \left(\frac{2.00 \times 37}{0.241}\right)$$
$$= 49.7 \text{ min.}$$

From equation (12.6),

$$\text{process time} = 49.7 - (0.4 \times 11)$$
$$= 45.3 \text{ min}$$

This gives the process time for $F_0 = 7$ min. If the process time had been given, it would be possible to reverse the calculation to find F_0.

Improved general (graphical) method

This method is based on the fact that different combinations of temperature and time have the same lethal effect on micro-organisms (Chapter 1). *Lethality* is therefore the integrated effect of temperature and time on micro-organisms. As the temperature increases, there is a logarithmic reduction in the time needed to destroy the same number of micro-organisms. This is expressed as the *lethal rate* (a dimensionless number that is the reciprocal of TDT) and is shown by the following equation:

$$\text{Lethal rate} = 10^{(\theta - 121)/z} \qquad 12.7$$

where θ (ºC) = temperature of heating.

The TDT at a given processing temperature is compared to a reference temperature (T) of 121ºC. For example, if a product is processed at 115ºC and the most heat-resistant micro-organism has a z value of 10ºC,

$$\text{Lethal rate} = 10^{(115-121)/10}$$
$$= 0.25$$

As the temperature of a food increases during processing, there is a higher rate of microbial destruction. The initial heating part of the process contributes little towards total lethality until the retort temperature is approached and most of the accumulated lethality takes place in the last few minutes, before cooling begins.

The lethal rate depends on the z value of the micro-organism on which the process is based and the product temperature, and tables of lethal rate values are available (Table 12.4 is for $z = 10$, the value for most spoilage micro-organisms). This method is preferable in practical situations for determining the impact of a process in terms of equivalent temperature/time relationships.

Sample problem 12.2

A convective heating food is sterilised at 115ºC to give F_0 = 7 min. The come-up time of retort is 11 min. Calculate the processing time from the following heat penetration data. Cooling started after 60 min.

Process time (min)	Temperature (ºC)	Process time (min)	Temperature (ºC)
0	95	35	115.5
5	101	40	115.5
10	108.5	45	115.5
15	111.5	50	115.6
20	113	55	115.6
25	115.5	60	115.6
30	115.5	65	100

Solution to Sample problem 12.2

Lethal rates can be found at selected points on a heat penetration curve either by constructing a TDT curve and taking the reciprocal of TDTs (from Fig. 1.14) at the selected temperatures, or by consulting the appropriate lethal rate table (Table 12.4). Lethal rates are then plotted against processing time (Fig. 12.5) and the area under the curve is measured by counting squares or using a planimeter.

Process time (min)	Lethal rate	Process time (min)	Lethal rate
0	0.002	35	0.218
5	0.01	40	0.218
10	0.055	45	0.224
15	0.109	50	0.224
20	0.155	55	0.224
25	0.218	60	0.224
30	0.218	65	0.005

For convection heating foods, the lethal rate curve is used to find the point in the process when heating should cease. A line is drawn parallel to the cooling part of the curve so that the total area enclosed by the curve is equal to the required lethality. The area under the curve ACE is 100.5 cm^2. As 1 cm^2 is 0.1 min at 121ºC, the area ACE represents 10.05 min at 121ºC. Therefore by reducing the area under the curve ABD to 70 cm^2 ($F = 7$ min), the process time is 45 min. Thus the process time required for $F_0 = 7$ min is 45 min.

Examples of F_0 values used commercially are shown in Table 12.5. With conduction heating foods the temperature at the centre of the container may continue to rise after cooling commences, because of the low rate of heat transfer. For these foods it is necessary to determine lethality after a number of trials in which heating is stopped at different times.

Table 12.4 Lethal rates for z = 10°C

Temperature (°C)	Lethal rate (min[a])	Temperature (°C)	Lethal rate (min)
90	0.001	108	0.049
92	0.001	110	0.077
94	0.002	112	0.123
96	0.003	114	0.195
98	0.005	116	0.308
100	0.008	118	0.489
102	0.012	120	0.774
104	0.019	122	1.227
106	0.031	124	1.945

[a] At 121°C per minute at θ_r.
Adapted from Stumbo (1973).

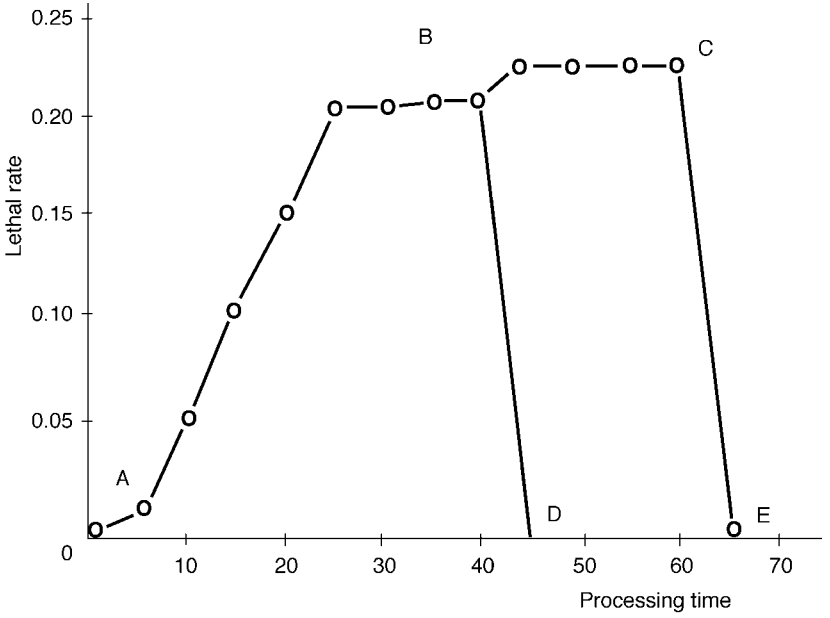


Fig. 12.5 Lethal rate curve.

Table 12.5 F_0 values for selected commercial processes

Product	F_0 values
Carrots in brine	3–4
Beans in tomato sauce	4–6
Herrings in tomato sauce	6–8
Meat in gravy	12–15

12.1.2 Retorting (heat processing)

The shelf life of sterilised foods depends in part on the ability of the container to isolate the food completely from the environment. The four major types of heat-sterilisable container are:

1. metal cans
2. glass jars or bottles
3. flexible pouches
4. rigid trays.

These materials are described in Chapter 24 and methods of filling and sealing are described in Chapter 25.

Before filled containers are processed, it is necessary to remove air by an operation termed 'exhausting'. This prevents air expanding with the heat and therefore reduces strain on the container. The removal of oxygen also prevents internal corrosion and oxidative changes in some foods. Steam replaces the air and on cooling forms a partial vacuum in the head space. Containers are exhausted by:

- hot filling the food into the container (commonly used as it also pre-heats food which reduces processing times)
- cold filling the food and then heating the container and contents to 80–95ºC with the lid partially sealed (clinched)
- mechanical removal of the air using a vacuum pump
- steam flow closing, where a blast of steam (at 34–41.5 $\times$ 10^3 Pa) carries air away from the surface of the food immediately before the container is sealed. This method is best suited to liquid foods where there is little air trapped in the product and the surface is flat and does not interrupt the flow of steam.

Heating by saturated steam

Latent heat is transferred to food when saturated steam condenses on the outside of the container. If air is trapped inside the retort, it forms an insulating boundary film (Chapter 1) around the cans which prevents the steam from condensing and causes under-processing of the food. It also produces a lower temperature than that obtained with saturated steam (Table 12.6). It is therefore important that all air is removed from the retort by the incoming steam using a procedure known as *venting*.

A major problem with processing solid or viscous foods is the low rate of heat penetration to the thermal centre. As a result, over-processing causes damage to nutritional and sensory characteristics of food near the walls of the container in addition to long processing times and hence low productivity. Methods that are used to increase the rate of heat transfer include the use of thinner profile containers and agitation of containers. An increase in retort temperature would also reduce processing times and protect nutritional and sensory qualities, but this is usually impractical; the higher pressures would require substantially stronger and hence more expensive containers and processing equipment.

After sterilisation the containers are cooled by sprays of water. Steam is rapidly condensed in the retort, but the food cools more slowly and the pressure in the containers remains high. Compressed air is therefore used to equalise the pressure to prevent strain on the container seams (pressure cooling). When the food has cooled to below 100ºC, the over-pressure of air is removed and cooling continues to approximately 40ºC. At this temperature, moisture on the container dries which prevents surface corrosion and allows label adhesives to set more rapidly.

Table 12.6 Temperatures of saturated steam at gauge pressures from 0 kPa to 199 kPa (0–29 lb ft^{-2})

Temperature (ºC)	Pressure (lb ft^{-2})	Pressure (kPa)	Temperature (ºC)	Pressure (lb ft^{-2})	Pressure (kPa)
100.0	0	0	121.0	15	103.4
101.9	1	6.9	122.0	16	110.3
103.6	2	13.8	123.0	17	117.2
105.3	3	20.7	124.1	18	124.1
106.9	4	27.6	125.0	19	131.0
108.4	5	34.5	126.0	20	137.9
109.8	6	41.4	126.9	21	144.8
111.3	7	48.3	127.9	22	151.7
112.6	8	55.2	128.7	23	158.6
113.9	9	62.1	129.6	24	165.5
115.2	10	68.9	130.4	25	172.4
116.4	11	75.8	131.2	26	179.3
117.6	12	82.7	132.1	27	186.2
118.8	13	89.6	133.0	28	193.1
119.9	14	96.5	133.6	29	199.9

Heating by hot water

Foods are processed in glass containers or flexible pouches under hot water with an overpressure of air. Glass containers are thicker than metal cans to provide adequate strength. This, together with the lower thermal conductivity of glass (Table 1.5), results in slower heat penetration and longer processing times than for cans and there is a higher risk of thermal shock to the container. Foods in rigid polymer trays or flexible pouches heat more rapidly owing to the thinner material and smaller cross-section of the container. This enables savings in energy and causes minimum overheating near the container wall. Liquid or semi-liquid foods are often processed horizontally to ensure that the thickness of food is constant across the pouch. Vertical packs promote better circulation of hot water in the retort, but special frames are necessary to prevent the pouches from bulging at the bottom, which would alter the rate of heat penetration and hence the degree of sterilisation achieved. Although pouches are popular in the Far East and for containing military rations, they have yet to achieve large-scale commercial use in Europe and the United States.

Heating by flames

Sterilisation at atmospheric pressure using direct flame heating of spinning cans is described by Beauvais *et al.* (1961), Casimir (1975) and Anon. (1984). High rates of heat transfer are possible at flame temperatures of 1770ºC. The consequent short processing times produce foods of high quality and reduce energy consumption by 20% compared with conventional canning. No brine or syrup is used in the can and smaller cans may be used which reduces transport costs by 20–30%. However, high internal pressures (275×10^3 Pa at 130ºC) limit this method to small cans. It is used, for example, to process mushrooms, sweetcorn, green beans, pears and cubed beef.

12.1.3 Equipment

Sterilising retorts may be batch or continuous in operation. Batch retorts may be vertical or horizontal; the latter are easier to load and unload and have facilities for agitating

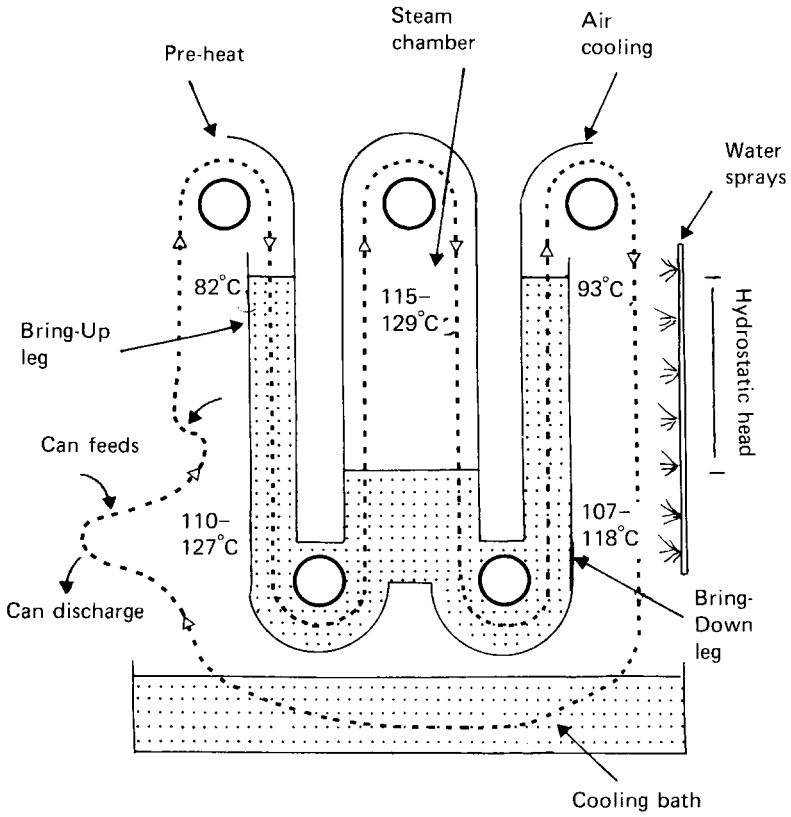


Fig. 12.6 Continuous hydrostatic steriliser.

containers, but require more floor space. For example, the 'Orbitort' consists of a pressure vessel that contains two concentric cages. Cans are loaded horizontally into the annular space between the cages and when full, the retort is sealed. The cages hold the cans against guide rails as they are slowly rotated to cause the headspace bubble to stir the contents (Brennan *et al.*, 1990).

Continuous retorts (for example, Fig 12.6) permit close control over the processing conditions and hence produce more uniform products. They produce gradual changes in pressure inside cans, and therefore less strain on the can seams compared with batch equipment. The main disadvantages include a high in-process stock which would be lost if a breakdown occurred, and in some, problems with metal corrosion and contamination by thermophilic bacteria if adequate preventative measures are not taken.

The main types are cooker-coolers, rotary sterilisers and hydrostatic sterilisers. Cooker-coolers carry cans on a conveyor through three sections of a tunnel that are maintained at different pressures for pre-heating, sterilising and cooling. Rotary sterilisers consist of a slowly rotating drum inside a pressure vessel. The vessel has a helical track on the inner wall and cans are moved along this by the drum, through three pressure sections separated by pressure locks. Again the contents are mixed as cans travel around the helix. Nair (1964) describes four types of hydrostatic steriliser in detail. In practice, large continuous sterilisers are used for the production of high-volume products

(for example, 1000 cans per min) where there is no requirement to regularly change the container size or processing conditions.

Computer control of retorts is described by Bown (1985, 1987), Hayakawa (1977), Holdsworth (1983) and Simpson *et al.* (1993). Selected process variables are monitored including:

- temperature of the raw material
- temperature of the cooling water
- temperature of steam
- time of processing
- heating and cooling rates.

This information is then processed by computer to calculate the accumulated lethality and to control the air, steam and water flow rates, to produce the required degree of sterility in the product with minimum energy expenditure. Further details of automatic process control are given in Chapter 2.

12.2 Ultra high-temperature (UHT)/aseptic processes

Higher processing temperatures for a shorter time are possible if the product is sterilised before it is filled into pre-sterilised containers in a sterile atmosphere. This forms the basis of UHT processing (also termed *aseptic processing*). It is used to sterilise a wide range of liquid foods, including milk, fruit juices and concentrates, cream, yoghurt, wine, salad dressing, egg and ice cream mix. It can also be used to process foods which contain small discrete particles, for example cottage cheese, baby foods, tomato products, fruit and vegetables, soups and rice desserts. Processes for larger-particulate foods have been developed (Manvell, 1987), and more recently, ohmic heating (Chapter 18) has been successfully used for UHT processing. The high quality of UHT foods competes well with chilled and frozen foods (Chapters 19 and 21), and UHT has an important additional advantage of a shelf life of at least six months without refrigeration.

The advantages of UHT processing compared with canning are summarised in Table 12.7. For example, UHT processing conditions are independent of container size. Whereas conventional retorting of A2 cans of vegetable soup requires 70 min at 121ºC to achieve an F_0 value of 7 min, followed by 50 min cooling, aseptic processing in a scraped-surface heat exchanger at 140ºC for 5 s gives an F_0 value of 9 min. Increasing the can size to A10 increases the processing time to 218 min, whereas with aseptic processing the sterilisation time is the same. This permits the use of very large containers (for example 1 tonne aseptic bags of tomato purée or liquid egg, used as an ingredient for other manufacturing processes).

The main limitations of UHT processing are the cost and complexity of the plant, which arises from the necessity to sterilise packaging materials, associated pipework and tanks, the maintenance of sterile air and surfaces in filling machines, and the higher skill levels required by operators and maintenance staff (Rose, 1986).

12.2.1 Theory

For a given increase in temperature, the rate of destruction of micro-organisms and many enzymes increases faster than the rate of destruction of nutrients and sensory components

Table 12.7 Comparison of conventional canning and aseptic processing and packaging

Criteria	Retorting	Aseptic processing and packaging
Product sterilisation		
Delivery	Unsteady state	Precise, isothermal
Process calculations		
Fluids	Routine – convection	Routine
Particulates	Routine – conduction or broken heating	Complex
Other sterilisation required	None	Complex (process equipment, containers, lids, aseptic tunnel)
Energy efficiency	Lower	30% saving or more
Sensory quality	Unsuited to heat sensitive foods	Superior – suitable for homogenous heat sensitive foods
Nutrient loss	High	Minimal
Value added	Lower	Higher
Convenience	Shelf stable	Shelf stable
Suitability for microwave heating	Glass and semi-rigid containers	All non-foil rigid and semi-rigid containers
Production rate	High (600–1000/min)	Medium (500/min)
Handling/labour costs	High	Low
Downtime	Minimal (mostly seamer and labeller)	Re-sterilisation needed if loss of sterility in filler or steriliser
Flexibility for different container sizes	Need different process delivery and/or retorts	Single filler for different container sizes
Survival of heat resistant enzymes	Rare	Common in some foods (e.g. milk)
Spoilage troubleshooting	Simple	
Low acid particulate processing	Routine	Not in practice (data from high acid systems being used to design low acid systems)
Post-process additions	Not possible	Possible to add filter sterilised enzymes or probiotics after heat processing

Adapted from David (1996).

(Fig. 12.7 and Chapter 1) and food quality is therefore better retained at higher processing temperatures. However some enzymes, for example proteases and lipases that are secreted by psychrotrophic micro-organisms in milk, are more heat resistant. These are not destroyed by some UHT treatments and may cause changes to the flavour of products during prolonged storage (David, 1996).

In contrast with in-container sterilisation where the most lethal effect frequently occurs at the end of the heating stage and the beginning of the cooling stage, UHT processes heat the food rapidly to a holding temperature and the major part of the lethality accumulates at a constant temperature. The sterilising value is calculated by multiplying the lethal rate at the holding temperature by the holding time. The come-up time and cooling periods are very short and in most cases are treated as a safety factor. The flow rate for the fastest moving particle and longest time needed for heat transfer from the liquid to the centre of the particle are together used to determine the time and temperature needed to achieve the required F_0 value. The criteria for UHT processing are therefore the same as for canning: that is the attainment of commercial sterility. Typical minimum time–temperature conditions needed to destroy *Cl. botulinum* ($F_0 = 3$) are 1.8 s at 141°C. In the UK the minimum heat treatment for dairy products are set out in regulations (1 s at 135°C for milk, 2 s at 140°C for cream and milk-based products and 2 s at 148.9°C for ice cream mixes) (Lewis, 1993).

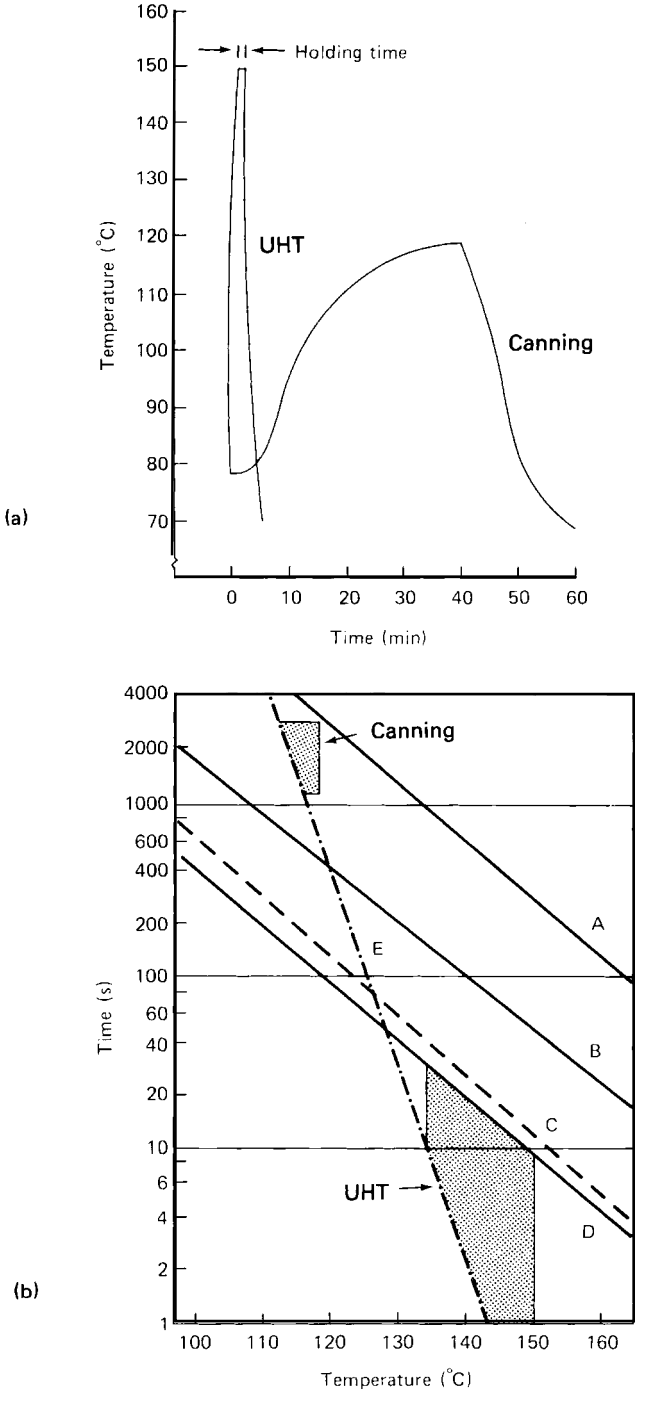


Fig. 12.7 (a) Time–temperature conditions for UHT and canning; (b) rates of microbial and nutrient destruction in UHT processing: line A, 40% thiamin; line B, 10% thiamin; line C, 1% lysine; line D, 3% thiamin; line E, microbial.
(After Killeit (1986).)

In addition to the use of F_0 to assess microbial destruction, a further two parameters are used in dairy UHT processing: the B^* value is used to measure the total integrated lethal effect of a process and the C^* value which measures the total chemical damage taking place during a process. The reference temperature used for these values is 135ºC. A process that is given a B^* value = 1 will result in a 9D reduction in spores ($z = 10.5$ºC) and would be equivalent to 10.1 s at 135ºC. Similarly a process given a C^* value = 1 will cause 3% loss of thiamine and would be equivalent to 30.5 s at 135ºC (Lewis, 1993).

Calculation of holding time uses equations (12.8) and (12.9):

$$B^* = 10^{(\theta-135)/10.5}.t/10.1 \quad [12.8]$$

$$C^* = 10^{(\theta-135)31.4}.t/30.5 \quad [12.9]$$

where θ (ºC) = processing temperature and t (s) = holding time.

Ideally a process should maximise B^* and minimise C^*, unless a specific chemical (for example an enzyme or natural toxin such as trypsin inhibitor) is to be destroyed or vegetable tissues are required to be softened.

12.2.2 Processing

Food is heated in relatively thin layers in a continuous heat exchanger with close control over the sterilisation temperature and holding time. It is important to know the shortest time that any particle can take to pass through the holding section and the rate of heat transfer from the liquid to the centre of the particle, to ensure that microbial spores cannot survive the process. The length of the holding tube is often calculated as twice the size of the average liquid flow rate (Ohlsson, 1992). It is also important to achieve turbulent flow (Chapter 1) if possible because the spread of residence times is smaller. With viscous foods, flow is likely to be streamline and result in a wider spread of residence times; the minimum may be only half the average time. The minimum time should therefore be greater than that specified for the product to avoid under-processing. Similarly there should be close control over the particle size range in particulate products. For example, if a process is designed to sterilise 14 mm particles to $F_0 = 6$, it can be calculated that the holding tube should be 13 m long. However, if a 20 mm particle passes through under these conditions, it will only reach $F_0 = 0.5$ and will thus be seriously under-processed. Conversely, a 10 mm diameter particle will reach $F_0 = 20$ and will be over-processed (Ohlsson, 1992).

The sterilised product is cooled in a second heat exchanger, or in a vacuum chamber if deaeration is also required. Containers are not required to withstand sterilisation conditions, and laminated cartons (Chapter 24) are therefore widely used. They have considerable economic advantages compared with cans and bottles, in both the cost of the pack and the transport and storage costs. Cartons are pre-sterilised with hydrogen peroxide, and filling machines are enclosed and maintained in a sterile condition by ultraviolet light and filtered air. A positive air pressure is maintained in the filling machine to prevent entry of contaminants (also Chapter 25).

The process is successfully applied to liquid and small-particulate foods but until recently, there were problems in processing larger pieces of solid food. The major difficulties were:

- enzyme inactivation at the centre of the pieces of food causes overcooking of the surfaces, thus limiting particle sizes

- agitation is necessary to improve the rate of heat transfer and to aid temperature distribution, but this causes damage to the product
- until recently there was a lack of suitable equipment for processing and filling
- settling of solids is a problem if the equipment has a holding tube. This causes uncontrolled and overlong holding times and variable proportions of solids in the filled product (Hersom, 1984).

These problems have now been addressed for larger particulates (up to 2.5 cm) using a number of processes, including the APV 'Jupiter' system, the Alfa-Laval 'Twintherm' system, the Stork 'single flow fraction specific thermal processing (FSTP)' system (Section 12.2.3) or ohmic heating (Chapter 18). Details of a protocol for validating the successful aseptic processing of particulates for the American Food and Drug Administration are given by Palaniappan and Sizer (1997). The flow of two-phase particulates through heat exchangers is highly complex and research is continuing to gain a better understanding and control of heat transfer in these systems. In particular the residence time distribution of each component should be known and the fastest particles used to monitor safety and the slowest used to monitor quality (Lewis, 1993).

12.2.3 Equipment

A theoretically ideal UHT process would heat the product instantly to the required temperature, hold it at that temperature to achieve sterility and cool it instantly to filling temperature. In practice the degree to which this is achieved depends in part on the method used to heat the food and in part on the sophistication of control and hence the cost of equipment. It also depends on the properties of the food (for example viscosity, presence of particles, heat sensitivity and tendency to form deposits on hot surfaces). With the exception of ohmic heating, equipment used for UHT processing has the following characteristics:

- operation above 132ºC
- exposure of a relatively small volume of product to a large surface area for heat transfer
- maintenance of turbulence in the product as it passes over the heating surface
- use of pumps to give a constant delivery of product against the pressure in the heat exchanger
- constant cleaning of the heating surfaces to maintain high rates of heat transfer and to reduce burning-on of the product. (The chemical composition and formation of deposits during UHT treatment of milk are described by Lalande *et al.* (1984).)

Equipment is classified according to the method of heating into:

- direct systems (steam injection and steam infusion)
- indirect systems (plate heat exchangers, tubular heat exchangers (concentric tube or shell-and-tube) and scraped surface heat exchangers)
- other systems (microwave, dielectric, ohmic and induction heating (Chapter 18)).

Direct methods

Details of equipment from different manufacturers are reviewed by Ramesh (1999).

Steam injection (uperisation) and steam infusion are each used to intimately combine the product with potable (culinary) steam. In steam injection, steam at a pressure of 965×10^3 Pa is introduced into a pre-heated liquid product in fine bubbles by a steam

injector and rapidly heats the product to 150ºC. After a suitable holding period (for example 2.5 s) the product is flash cooled in a vacuum chamber to 70ºC, and condensed steam and volatiles in the product are removed. The moisture content of the product therefore returns to approximately the same level as the raw material.

The main advantages of this system are as follows:

- it is one of the fastest methods of heating and the fastest method of cooling and is therefore suitable for more heat-sensitive foods
- volatile removal is an advantage with some foods (for example milk).

However, there are limitations:

- the method is only suitable for low-viscosity products
- there is relatively poor control over processing conditions
- there is a requirement for potable steam which is more expensive to produce than normal processing steam
- regeneration of energy is less than 50% compared with more than 90% in indirect systems
- flexibility for changing to different types of product is low.

Further details are given by Burton and Perkins (1970), Zadow (1975), Burton (1988), Lewis and Heppel (2000) and Carlson (1996).

In steam infusion the food is sprayed in a free-falling film into high-pressure (450×10^3 Pa) potable steam in a pressurised vessel. It is heated to 142–146ºC in 0.3 s, and is held for 3 s in a holding tube before flash cooling in a vacuum chamber to 65–70ºC. Heat from the flash cooling is used to pre-heat the feed material.

In both systems microcomputers control temperature, pressure, level, flow rate, valve operation and the cleaning sequence, at production rates of up to 9000 kg h^{-1} (Swientek, 1983). Steam infusion has advantages over injection methods because the liquid does not contact hotter surfaces and burning-on is therefore reduced. Other advantages include:

- almost instantaneous heating of the food to the temperature of the steam, and very rapid cooling which results in high retention of sensory characteristics and nutritional properties
- greater control over processing conditions than steam injection
- lower risk of localised overheating of the product
- the method is more suitable for higher viscosity foods compared to steam injection.

The main disadvantages, in addition to the disadvantages of steam injection, are blockage of the spray nozzles and separation of components in some foods. Further details are given by Burton (1988).

Indirect systems

A description of the historical development of different types of heat exchangers for indirect UHT heating is given by Carlson (1996). *Plate heat exchangers* are described in detail for pasteurisation (Chapter 11). In UHT sterilisation, they have a number of limitations which are due to the higher temperatures and pressures involved (Table 12.8). *Tube and shell heat exchangers* are described in Chapter 13, in their application to evaporation by boiling. The advantages and limitations of this type of equipment for UHT processing are described in Table 12.8.

There are a large number of designs of tube and shell heat exchangers, including the *concentric tube heat exchanger* which is a combination of both plate and tubular designs

Table 12.8 Comparison of plate and tube-and-shell heat exchangers for UHT processing

Plate heat exchanger		Tube-and-shell heat exchanger	
Advantages	Limitations	Advantages	Limitations
• Relatively inexpensive • Economical in floor space and water consumption • Efficient in energy use (> 90% energy regeneration) • Flexible changes to production rate, by varying the number of plates • Easily inspected by opening the plate stack	• Operating pressures limited by the plate gaskets to approximately 700 kPa • Liquid velocities at relatively low pressure also low ($1.5-2\,m\,s^{-1}$) • Low flow rates can cause uneven heating and solids deposits on the plates which require more frequent cleaning • Gaskets susceptible to high temperatures and caustic cleaning fluids and are replaced more regularly than in pasteurisation • Limited to low viscosity liquids (up to $1.5\,N\,s\,m^{-2}$) • Careful initial sterilisation of the large mass of metal in the plate stack is necessary for uniform expansion to prevent distortion and damage to plates or seals • Liable to fouling	• Few seals and easier cleaning and maintenance of aseptic conditions • Operation at higher pressures (7000–10 000 kPa) and higher liquid flow rates ($6\,m\,s^{-1}$) than plate heat exchangers • Turbulent flow at tube walls due to higher flow rates • Hence more uniform heat transfer and less product deposition	• Difficulty in inspecting heat transfer surfaces for food deposits • Limited to relatively low-viscosity foods (up to $1.5\,N\,s\,m^{-2}$) • Lower flexibility to changes in production capacity • Larger-diameter tubes cannot be used because higher pressures needed to maintain the liquid velocity and large-diameter pipes have a lower resistance to pressure • Any increase in production rate requires duplication of the equipment

(Fig. 12.8(a)). Counter-current flow and helical corrugations are used to generate turbulence and hence to increase the rate of heat transfer. This equipment is able to operate at high pressures (up to 2000×10^3 Pa) with viscous liquids. Turbulence can be further increased at the expense of throughput or maximum particle size, by altering the relative positions of tubes (Fig. 12.8(b)).

Scraped-surface heat exchangers are also used for freezing (Chapter 21), for evaporation by boiling (Chapter 13) and for the continuous production of margarine and butter (Chapter 4). Their main advantages for UHT processing are their suitability for viscous foods and particulates (less than 1 cm), and their flexibility for different products by changing the geometry of the rotor assembly. However, they have high capital and operating costs and heat recovery is not possible. They are used to process fruit sauces and fruit bases for yoghurts and pies.

During the 1970s and 1980s, systems were developed that involve separate treatment of the liquid and particulate components of a food to enable greater control over the F_0 values, particularly for sterilisation of the particles. For example, the *'Jupiter' double-cone heat exchanger* combines indirect heating in a jacketed double cone, with direct heating by steam or superheated liquor. It has been used for processing large (2–2.5 cm) particles of food. In a sequence of microprocessor-controlled operations, solid pieces of

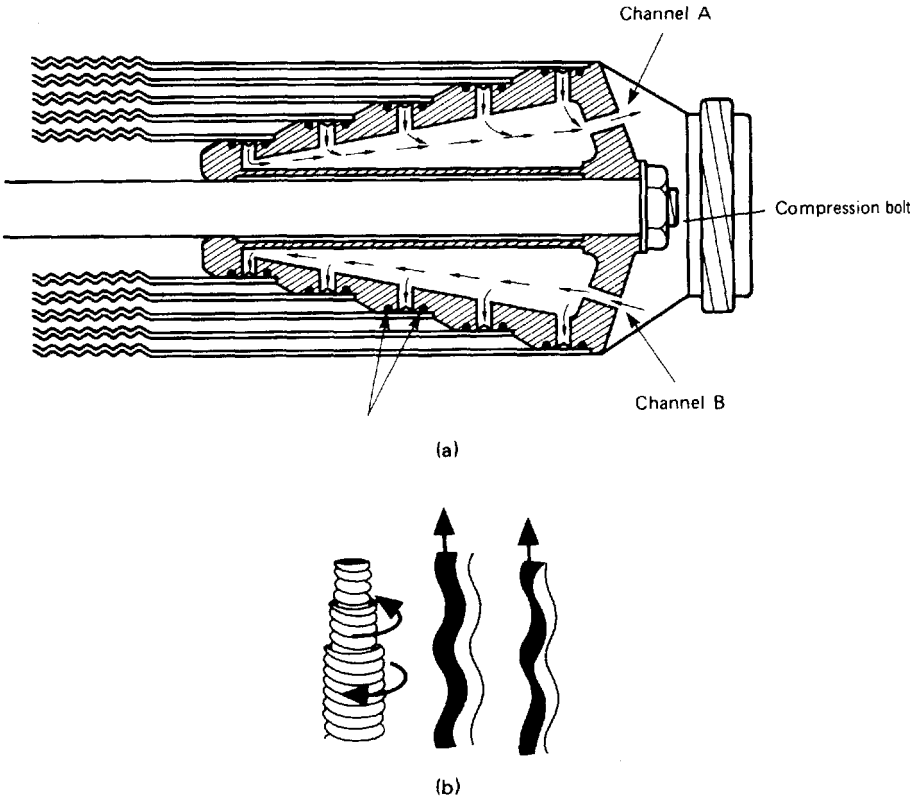


Fig. 12.8 Concentric tube heat exchanger: (a) cross-section; (b) effect of rotation of tubes. (Courtesy of HRS Heat Exchangers Ltd.)

food are fed into a double-cone vessel, which is then rotated slowly (4–12 rpm) on a horizontal axis. Steam at 206×10^3 Pa is introduced and the product is tumbled through the steam. Steam in the jacket is at the same temperature (for example 137°C) to prevent the food from burning onto the cone. Liquor, sterilised separately, is added during sterilisation to prevent damage to the solids by the tumbling action. Sterilisation times are for example 2–4 min to achieve $F_0 = 10$ for carrot cubes, depending on their size. After sterilisation the product is rapidly cooled with cold water and sterile air, and the condensate-water-stock is removed. The liquid portion of the product is sterilised separately in a plate or tubular system and added to the solids. The cone then acts as a mixer. The blended solids–liquids are discharged to an aseptic filler using an over-pressure of sterile air. This avoids pumping the softened product and further reduces damage to the food. Cooking liquor from the solids is used to make sauce to top up containers or to inject into solids during subsequent processing. However, the Jupiter has a relatively low capacity, complex operation and relatively high capital costs, and for these reasons has not been widely adopted.

The 'Twintherm' system is a semi-continuous process in which particulate food is heated by direct steam injection in a pressurised, horizontal, cylindrical vessel that rotates slowly. Once the particles have been held for the required time to achieve F_0 values, they are cooled evaporatively and liquid, which has been sterilised in conventional UHT heat exchangers, is used to carry the particles to an aseptic filler. It is claimed to allow more uniform and gentle treatment of particles, compared to continuous processes and it is

used commercially to produce soups for the European market (Ohlsson, 1992; Alkskog, 1991).

The single flow FSTP system employs a separate holding section which is a cylindrical vessel containing slowly rotating fork blades on a central shaft. These blades form cages which hold the particles as they are rotated around the cylinder from the inlet to discharge pipes. Liquid moves freely through the cages, giving rapid heat transfer. Different holding sections can be used for different particle sizes which reduces the risk of over-processing of smaller particles that is found in more traditional systems. Again the liquid component is sterilised separately in conventional tubular heat exchangers and is then used to carry the sterilised particles to the filler. The system is described in detail by Hermans (1991).

More recent developments, described by Carlson (1996) and Lewis and Heppel (2000), include *tube-in-tube heat exchangers* which are a development of plate heat exchangers, in which the thin-walled plates are formed into a tube, and corrugations or twists in the tubes are introduced to promote turbulent flow. One tube is inserted inside another, which may be repeated two or more times and heat transfer takes place across the tube walls. Incoming material is heated by sterilised product to regenerate heat and increase energy efficiency. Steam from hot water is used for the final heating and after initial cooling by the incoming material, the product is cooled with cold water. This system is relatively low cost and is widely used, although it suffers from some drawbacks: seal integrity is critical to prevent mixing of incoming material, product or heating and cooling media; the seal strength limits the pressure that can be used and seals may be difficult to clean. Additionally, the equipment is restricted to relatively low viscosity liquids that do not cause significant fouling (Carlson, 1996), but has been widely used for fruit juices, milk and dairy products.

The problems encountered in straight tube heat exchangers described above are largely overcome by forming a single tube into a continuous helix or coil, which has a carefully defined ratio between the diameter of the coil and the diameter of the tube. The coil is contained within an insulating material to minimise heat losses. The design of the coil promotes secondary flow of liquid within the tube, and this causes turbulence at relatively low flow rates, and high rates of heat transfer (between two and four times the rates in tube-in-tube or shell-and-tube heat exchangers (Carlson, 1996). This enables processing of heat sensitive products (e.g. liquid egg) and products that cause fouling of heat exchanger surfaces. The mixing action in the coil gives a uniform distribution of particles, making the equipment suitable for salad dressings, fruit purées and other foods that contain a range of particle sizes, as well as for high viscosity liquids such as cheese sauce. Additionally, the continuous tube has no seals and is easily cleaned-in-place, and the simple design is virtually maintenance-free.

Other systems

Three other systems have been developed: the ‘Multi-therm’, ‘Achilles’ and ohmic heating. In the first two the food is heated by a combination of hot liquid and microwave energy. In ohmic heating a conducting fluid is heated directly by electrical energy (Chapter 18). Conversion efficiencies from electrical energy to heat of greater than 90% are claimed, and particulate feeds may be processed without shearing forces associated with some other types of heat exchangers. Indirect systems are automatically cleaned in place after 3–4 h of operation to remove accumulated deposits. The cleaning programme does not involve loss of sterile conditions, and processing resumes immediately afterwards.

12.3 Effect on foods

The purpose of heat sterilisation is to extend the shelf life of foods for storage at ambient temperatures, while minimising the changes in nutritional value and eating quality. Differences between the *D* and *z* values of micro-organisms, enzymes and sensory or nutritional components of foods are exploited to optimise processes for the retention of nutritional and sensory qualities (Swartzel, 1982). Preceding sections describe how this is achieved in practice by a reduction in size or cross-sectional area of containers, by agitation during processing or by aseptic processing. In this section the changes to foods caused by traditional canning techniques are compared with those caused by UHT processing.

12.3.1 Colour

The time–temperature combinations used in canning have a substantial effect on most naturally occurring pigments in foods. For example, in meats the red oxymyoglobin pigment is converted to brown metmyoglobin, and purplish myoglobin is converted to red–brown myohaemichromogen. Maillard browning and caramelisation also contribute to the colour of sterilised meats. However, this is an acceptable change in cooked meats. Sodium nitrite and sodium nitrate are added to some meat products to reduce the risk of growth of *C. botulinum*. The resulting red–pink coloration is due to nitric oxide myoglobin and metmyoglobin nitrite.

In fruits and vegetables, chlorophyll is converted to pheophytin, carotenoids are isomerised from 5, 6-epoxides to less intensely coloured 5, 8-epoxides, and anthocyanins are degraded to brown pigments. This loss of colour is often corrected using permitted synthetic colourants (Appendix C). Discolouration of canned foods during storage occurs, for example, when iron or tin react with anthocyanins to form a purple pigment, or when colourless leucoanthocyanins form pink anthocyanin complexes in some varieties of pears and quinces. In sterilised milk slight colour changes are due to caramelisation, Maillard browning and changes in the reflectivity of casein micelles.

In UHT processing, meat pigments change colour, but there is little caramelisation or Maillard browning. Carotenes and betanin are virtually unaffected, and chlorophyll and anthocyanins are better retained. There is an increase in whiteness in the colour of milk.

12.3.2 Flavour and aroma

In canned meats there are complex changes, for example pyrolysis, deamination and decarboxylation of amino acids, degradation, Maillard reactions and caramelisation of carbohydrates to furfural and hydroxymethylfurfural, and oxidation and decarboxylation of lipids. Interactions between these components produce more than 600 flavour compounds in ten chemical classes. In fruits and vegetables, changes are due to complex reactions which involve the degradation, recombination and volatilisation of aldehydes, ketones, sugars, lactones, amino acids and organic acids. In milk the development of a cooked flavour is due to denaturation of whey proteins to form hydrogen sulphide and the formation of lactones and methyl ketones from lipids. In aseptically sterilised foods the changes are again less severe, and the natural flavours of milk, fruit juices and vegetables are better retained. Changes to milk are discussed in detail by Burton (1988).

12.3.3 Texture or viscosity

In canned meats, changes in texture are caused by coagulation and a loss of waterholding capacity of proteins, which produces shrinkage and stiffening of muscle tissues. Softening is caused by hydrolysis of collagen, solubilisation of the resulting gelatin, and melting and dispersion of fats through the product. Polyphosphates (Appendix C) are added to some products to bind water. This increases the tenderness of the product and reduces shrinkage. In fruits and vegetables, softening is caused by hydrolysis of pectic materials, gelatinisation of starches and partial solubilisation of hemicelluloses, combined with a loss of cell turgor. Calcium salts may be added to blancher water (Chapter 10), or to brine or syrup, to form insoluble calcium pectate and thus to increase the firmness of the canned product. Different salts are needed for different types of fruit (for example calcium hydroxide for cherries, calcium chloride for tomatoes and calcium lactate for apples) owing to differences in the proportion of demethylated pectin in each product. Small changes in the viscosity of milk are caused by modification of K-casein, leading to an increased sensitivity to calcium precipitation and coagulation.

In aseptically processed milk and fruit juices the viscosity is unchanged. The texture of solid fruit and vegetable pieces is softer than the unprocessed food due to solubilisation of pectic materials and a loss of cell turgor but is considerably firmer than canned products. The relatively long time required for collagen hydrolysis and the relatively low temperature needed to prevent toughening of meat fibres are conditions found in canning but not in UHT processing. Toughening of meat is therefore likely under UHT conditions. The texture of meat purées is determined by size reduction and blending operations (Chapters 4 and 5) and is not substantially affected by aseptic processing.

Table 12.9 Loss of vitamins in canned and bottled foods (including losses due to preparation and blanching)

	Loss (%)											
Food	Carotene	Thiamin	Riboflavin	Niacin	Vitamin C	Panto-thenic acid	Vitamin B_6	Folacin	Bio-tin
Low-acid foods									
Carrots	0–9 (6)	67	38–60	32	75	54	80	59	40
Beef	—	67	100	100	—	—	—	—	—
Green beans	22–52	62	54–63	40	79	61	50	57	—
Mackerel	4	60	39	29	—	—	46	—	—
Milk	0	35	0	0	50–90	0	50	10–20	—
Mushrooms	—	80	46	52	33	54	—	84	54
Peas	0–30 (3)	75 (84)	47 (67)	71 (91)	67 (80)	80	69	59	78
Potatoes	—	56	44	56	28	—	59	—	—
Salmon	9	73	0	0	—	58	57	—	—
Spinach	0–32 (9)	80 (84)	45 (47)	50 (50)	72 (79)	78	75	35	67
Tomatoes	0 (2)	17 (22)	25 (59)	0 (1)	26 (26)	30	10	54	55
Acid foods									
Apple	0–4	31	48	—	74	15	0		
Cherries (sweet)	41	57	64	46	68	—	6		
Peaches	65 (70)	49 (57)	39	39 (38)	56 (58)	71	21		
Pears	—	45	45	0	73	69	18		
Pineapple	25	7 (10)	30	0	57 (57)	12	—		

The values in parentheses indicate the vitamin loss after storage for 12 months at 10–15ºC.
Adapted from De Ritter (1982), Rolls (1982), Burger (1982) and March (1982).

Table 12.10 Changes in nutritive value of milk after UHT and in-bottle sterilisation

Nutrient	Loss (%) on processing	
	UHT	In-bottle
Thiamin	10	35
Ascorbic acid	25	90
Vitamin B_{12}	10	90
Folic acid	10	50
Pantothenic acid	0	0
Biotin	0	0
β-carotene	0	0
Pyridoxine	10	50
Vitamin D	0	0
Whey proteins (denaturation)	12–40[a]	87
Lysine	–	10
Cystine	–	13
Biological value	–	6

[a] Direct UHT at 135ºC for 2 s (12.3%) and indirect UHT at 135ºC for 2 s (40.3%).
Adapted from Rolls (1982), Kiesker (1972) and Ford *et al*. (1969).

12.3.4 Nutritional value

Canning causes the hydrolysis of carbohydrates and lipids, but these nutrients remain available and the nutritional value of the food is not affected. Proteins are coagulated and, in canned meats, losses of amino acids are 10–20%. Reductions in lysine content are proportional to the severity of heating but rarely exceed 25%. The loss of tryptophan and, to a lesser extent, methionine, reduces the biological value of the proteins by 6–9%. Vitamin losses (Table 12.9) are mostly confined to thiamin (50–75%) and pantothenic acid (20–35%). In canned fruits and vegetables, significant losses may occur in all water-soluble vitamins, particularly ascorbic acid. However, there are large variations owing to differences in the types of food, the presence of residual oxygen in the container, and methods of preparation (peeling and slicing) or blanching (Chapters 3 and 10). In some foods, vitamins are transferred into the brine or syrup, which is also consumed. There is thus a smaller nutritional loss.

Sterilised soya–meat products may show an increase in nutritional value owing to a factor that decreases the stability of the trypsin inhibitor in soy beans. Aseptically processed meat and vegetable products lose thiamin and pyridoxine but other vitamins are largely unaffected. There are negligible vitamin losses in aseptically processed milk (Table 12.10) and lipids, carbohydrates and minerals are virtually unaffected. Riboflavin, pantothenic acid, biotin, nicotinic acid and vitamin B_6 are unaffected. The effect of processing conditions on vitamin content of canned foods are discussed in detail by Lamb *et al.* (1982), and changes in sterilised milk are discussed by Burton (1988). Ramesh (1999) has reviewed vitamin losses during UHT processing. Nutrient losses also occur during periods of prolonged storage, and these should also be considered when assessing the importance of sterilised foods in the diet.

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